Systematic Investment Strategies

Homework 5

Antonacci dual momentum, updated through 2026-06-30

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Summary of answers

(a) Data. Monthly total-return series, month-end, 1998-11-30 to 2026-06-30 (332 holding months after a 12-month lookback). Four modules, each with two risky assets and the same T-bill safe haven: equities VFINX/VGTSX, credit VWEHX/VFICX, real estate VGSIX/NLY, stress World Bank gold/VUSTX. T-bills are Kenneth French's monthly RF (1-month bill return), the closest public analogue of the paper's BofA Merrill Lynch 3-month T-bill index. Sources, cleaning, and the NLY/gold caveats are in §1.

(b) Implementation. Dual momentum as in Antonacci (2013): 12-month compounded total return with *no* skip-a-month (the paper's explicit choice), relative winner inside each module, then a T-bill hurdle for absolute momentum. Signal at month-end t, held during t+1. Composite is 25% each module, rebalanced monthly. Full-sample composite: CAGR 9.01638%, vol 7.44579%, Sharpe 0.93231, max DD -14.98158%.

(c) vs the paper. On the assignment's 1/00–12/09 window the four modules are in the same ballpark as Table 8, especially REITs (21.47342% vs paper 22.14000%). The composite earns 11.95796% with vol 7.43274% and Sharpe 1.19126. True out-of-sample (2012-01 to 2026-06) the composite slows to CAGR 6.99688%, Sharpe 0.75767, and lags US equities (15.01906%) and even 60/40 (9.81373%). Absolute momentum still does what it is for: GFC total return -0.91400% vs US -45.82052%. It does not add return in a long equity bull, and it is too slow for COVID (a two-month crash inside a positive 12-month trend).

(d) Extra credit. Two changes, frozen before looking at 2012–2026. Skip-last-month 12-1 dual, 1/N: OOS CAGR 9.00196% and Sharpe 0.97598 vs the paper rule's 6.99688% / 0.75767. Inverse-vol module weights make it worse (OOS CAGR 5.99819%) because they load 46.851% on the low-vol credit sleeve and starve the high-vol modules the paper wanted. 12-1 plus inverse-vol is in between (CAGR 8.12312%, max DD -10.74428%). The improvement that survives is 12-1, not risk parity.

1. Data

The paper's empirical work uses institutional total-return indices from January 1974 through December 2011 (Antonacci 2013, §2). Those series are not public. I map each paper index to a live total-return vehicle that (i) starts before 2011, (ii) covers 2007–09, (iii) runs through 2026-06-30, and (iv) is downloadable by a third party. Mutual-fund adjusted closes beat ETFs on history. All risky-asset returns are Yahoo Finance *auto-adjusted* closes (splits and dividends), sampled at calendar month-end (last available print). No price interpolation. No expense-ratio add-back: live vehicles already embed fees, so the update is net of the costs an implementer would have paid, unlike the paper (no transaction costs, institutional index levels).

Table 1: Mapping from Antonacci (2013) §2 to public total-return series.

Module	Role	Paper series	Series used	History / note
Equities	US large cap	MSCI US	VFINX	1976 / 1980 on Yahoo
Equities	World ex-US	EAFE+ (EAFE then ACWI ex-US)	VGTSX	Apr 1996; includes EM
Credit	High yield	BofA ML US Cash Pay HY	VWEHX	1978 / 1980 on Yahoo
Credit	Intermediate IG	Barclays US Intermediate Credit	VFICX	Oct 1993
Real estate	Equity REITs	FTSE NAREIT equity REITs	VGSIX	May 1996
Real estate	Mortgage REITs	FTSE NAREIT mortgage REITs	NLY	Oct 1997; single name
Stress	Gold	London PM gold fix	WB Pink Sheet gold	monthly average USD/oz
Stress	Long Treasuries	Barclays US Long Treasury	VUSTX	May 1986
Safe haven	T-bills	BofA ML 3-month T-bill index	French RF	monthly 1m T-bill return
Benchmark	US 60/40	World 60/40 analogue	0.6 VFINX + 0.4 VBMFX	monthly rebalanced

1.1 Why these pairs

Each module needs two risky assets that are different views of the same risk premium, plus one non-risky safe haven (assignment (a)). Equities: US vs rest-of-world, the paper's pair; VGTSX is ACWI ex-US rather than EAFE, so after 1987 it is closer to the paper's EAFE+ than EFA would be. Credit: high yield vs intermediate investment-grade, matching Table 3's duration-matched credit-risk contrast (HY vol comes from default risk, not duration). Real estate: equity REITs vs a mortgage REIT. Stress: gold (flight from uncertainty) vs long Treasuries (flight to quality). The same T-bill is the absolute-momentum hurdle and the cash sleeve in every module, as the paper does and the assignment allows.

1.2 Two imperfect mappings, stated plainly

Mortgage REITs. The paper uses the NAREIT mortgage-REIT *index*. The iShares mREIT ETF REM starts May 2007 as a fund but the liquid history that Yahoo returns cleanly from 2011, which would drop the GFC. I use Annaly (NLY), the largest agency mREIT, from October 1997. That is a single leveraged name, not an index: idiosyncratic rate, prepayment, and dividend-cut risk sit in the REIT module. On 2000–09 the dual REIT sleeve still landed close to Table 8 (21.47342% vs 22.14000%), which is the check I care about, but the mapping is the weakest in the panel.

Gold. FRED's LBMA PM/AM fix series (GOLDPMGBD228NLBM / GOLDAMGBD228NLBM) were pulled in January 2022 with the rest of the ICE Benchmark Administration files; the old fredgraph.csv URLs 404. I use the World Bank Pink Sheet monthly average gold price in USD/oz (public-domain compilation, datasets/gold-prices, through 2026-10). That is a *monthly average*, not the paper's month-end London PM print, so gold's measured vol is slightly smoothed. Timing is still valid: the June average is known at June month-end and is used only for the July holding. Gold pays no dividend; percent changes in the USD/oz series are the total return, as in the paper.

T-bills. French RF is a true monthly T-bill *return*, not a yield. I prefer it to converting FRED TB3MS with (1+y)^(1/12)−1. French RF is a one-month bill; the paper used a three-month bill index. For a 12-month ranking the difference is second-order. French RF in this file runs through 2026-09, so no yield conversion is needed inside the 2026-06 sample.

1.3 Sample

The binding start is NLY (1997-10-31). Twelve monthly returns plus a one-month implementation lag give a first holding month of 1998-11-30. Last holding month is 2026-06-30 (332 months, 27.66667 years). That covers the GFC, the assignment's 1/00–12/09 subsample, overlap with the paper through 2011-12, and 14.5 years of true out-of-sample. Processed prices and returns are in `data/processed/`; a third party does not need Yahoo.

2. Dual momentum, as implemented

Antonacci's two-stage rule, applied independently in each module at month-end *t*, after R_t is known:

1. **Relative momentum.** Among the module's two risky assets, pick the one with the higher trailing 12-month compounded total return $m_{i,t} = \Pi_{k=0..11} (1+R_{i,t-k}) - 1$.

2. **Absolute momentum.** If that winner also beat T-bills over the same 12 months, hold the winner during month *t*+1. Otherwise hold T-bills during *t*+1.

3. **Composite.** Equal 25% in each of the four module returns, rebalanced every month. Antonacci (footnote 7) cites DeMiguel, Garlappi and Uppal (2009): 1/N beats estimated optimal weights out of sample. I do not re-optimize.

The paper is explicit (p. 4) that it does *not* skip the last month, even though equity-momentum papers usually do, because the universe mixes gold, bonds and REITs. That is the baseline. The assignment's parenthetical "(without last-month reversal effects)" is the 12-1 alternative, run as extra credit in §5. Look-ahead is barred by construction: the holding during month *t* equals the choice made at *t*–1. Five unit tests in `tests/test_lookahead.py` pin down the lag, the skip, the T-bill overlay, and the inverse-vol lag.

I also compute the paper's three natural comparators (Table 14): relative-only (always hold the winner), absolute-only (each of the eight assets vs T-bills, then 1/8), and no-momentum 1/8 of the eight risky assets. 60/40 is 60% VFINX + 40% VBMFX, monthly. Section 12 factor alphas are not run.

2.1 Statistics

CAGR is geometric, (product of (1+R))^(12/n) − 1. Vol is sample monthly standard deviation × √12 (ddof=1). Sharpe is mean excess monthly return over T-bills, divided by the sample standard deviation of that excess, times √12 — the usual arithmetic Sharpe. I also store (CAGR − rf CAGR)/vol as sharpe_geom, which is closer to how practitioner tables sometimes look; headline Sharpes in this write-up are the arithmetic ones. Max drawdown is peak-to-trough on the month-end wealth index. % profit months is the fraction of months with R>0. Trades/year count holding *changes* after the first valid month, divided by years. Kurtosis is Pearson (normal = 3), matching Table 10.

3. Overlap with the paper

3.1 Table 8, January 2000 – December 2009

This is the subsample the assignment flags if the data reach 2007. Ours do. The paper's decade table is the cleanest apples-to-apples check, because it does not require 1974 history.

Table 2: Dual-momentum modules, Jan 2000 – Dec 2009, vs Antonacci Table 8. Paper Sharpe methodology is not fully specified; ours is arithmetic excess/vol. Composite is not in Table 8.

Module	Paper CAGR	Ours CAGR	Paper vol	Ours vol	Paper Sh.	Ours Sh.	Paper DD	Ours DD
Equities dual	9.49000%	7.41455%	9.41000%	9.71629%	0.39000	0.50365	-14.98000%	-19.53123%
Credit dual	7.62000%	5.81802%	4.39000%	4.57451%	0.45000	0.66014	-7.82000%	-10.91063%
REIT dual	22.14000%	21.47342%	16.45000%	19.98190%	0.90000	0.94500	-23.74000%	-28.03543%
Stress dual	13.27000%	11.00119%	16.49000%	13.89609%	0.43000	0.62812	-24.78000%	-24.48106%
Composite (ours)	—	11.95796%	—	7.43274%	—	1.19126	—	-13.15117%

Directionally the paper's ranking survives: REITs are the high-return high-vol sleeve (21.47342% vs paper 22.14000%), credit is the low-vol Sharpe engine, equities and stress sit in between. Every sleeve is a bit weaker than the paper, which is what you should expect from live mutual funds with fees vs institutional indices, a single-name mREIT, and monthly-average gold. The gaps are 1.8–2.3 percentage points of CAGR except REITs (–0.67 pp). I would not call this a replication of the 1974–2011 Table 10 composite (14.93% / 1.07 Sharpe) — different start date, different assets — but it is close enough on the requested decade that the *rule* is the same rule. Our 2000–09 composite did 11.95796% at 7.43274% vol, Sharpe 1.19126, max DD -13.15117%.

3.2 Utilization vs Table 11

Paper, 1974–2011 (Table 11): equities spent 22.6% of months in T-bills, credit 25.2%, REITs 26.3%, stress 17.8%. Full-sample utilization here:

Table 3: Fraction of months each asset is held, dual 12-0, full sample.

Module	Risky A (US / HY / EREIT / GOLD)	Risky B (INTL / IG / MREIT / TREAS)	T-bills
Equities	47.892%	31.024%	21.084%
Credit	50.602%	33.735%	15.663%
REITs	46.687%	37.651%	15.663%
Stress	53.614%	30.120%	16.265%

T-bill occupancy is similar to the paper, a bit lower (15.7–21.1% vs 17.8–26.3%). The mix inside the risky sleeves is more US- and gold-heavy than 1974–2011, which is the post-1998 sample, not a bug: US beat international for most of 2010–2026, and gold had two long bulls. All four modules sat in T-bills on 2.711% of months (paper 3.5%); two or more were in T-bills on 13.253% of months.

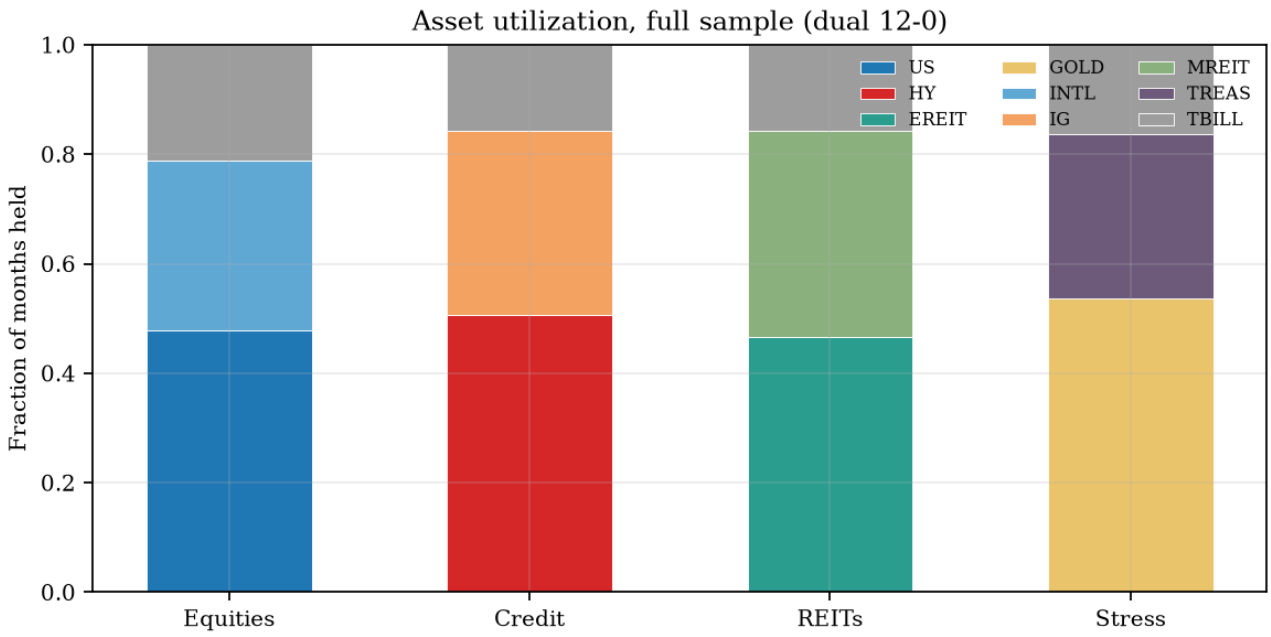


Figure 1: Asset utilization by module, full sample.

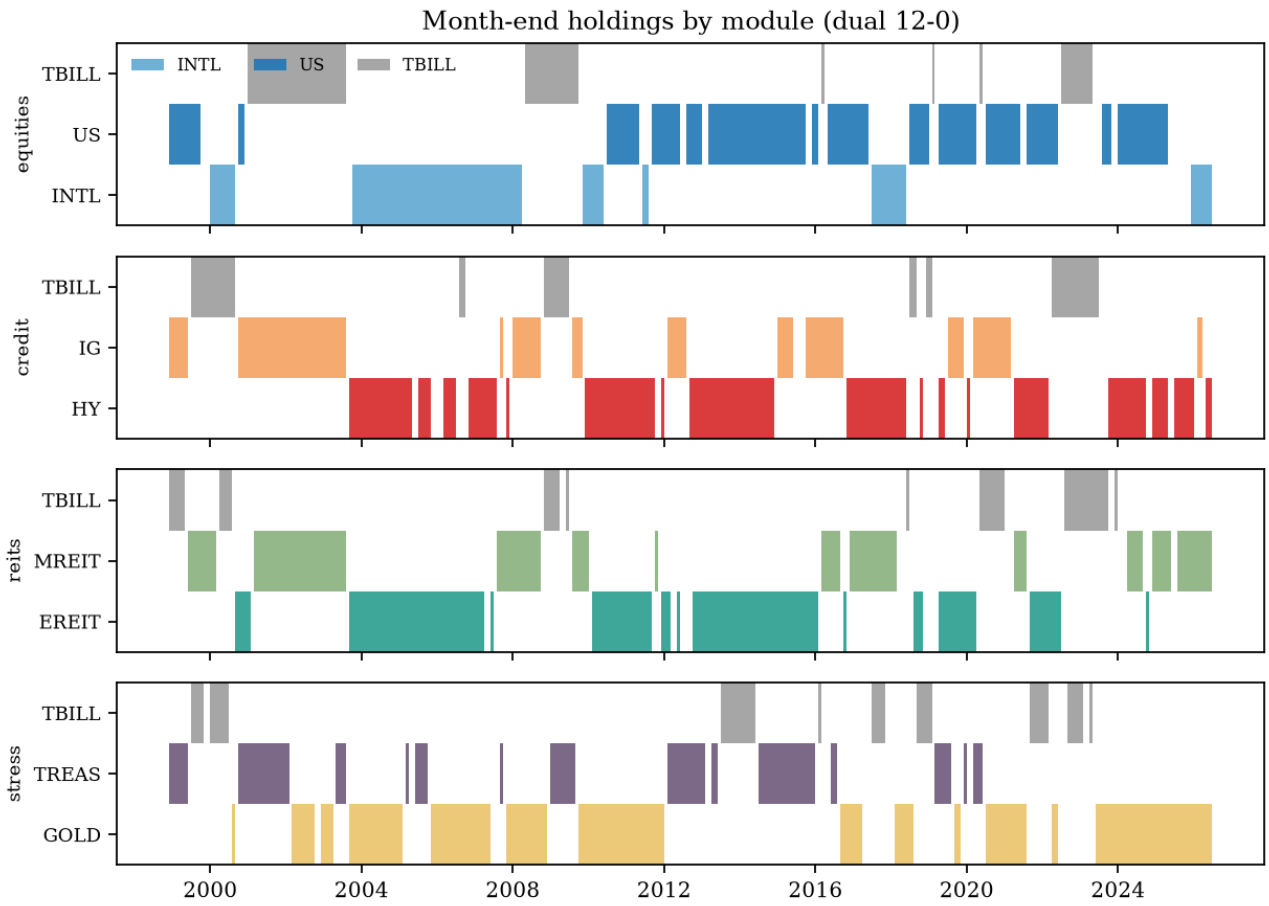


Figure 2: Month-end holdings. Grey = T-bills. The 2008 and 2022 cash spells are visible in equities; gold dominates the stress module after 2023.

3.3 The GFC, which is why absolute momentum exists

October 2007 – March 2009, total (not annualized) return: dual composite -0.91400%, relative-only composite -11.84940%, 1/8 risky -19.87949%, 60/40 -27.63502%, US equities -45.82052%. Dual lost less than one percent over the GFC window; US lost almost half. That is Table 1 / Figure 1 of the paper, on public funds. The gap between dual and relative-only (-0.91400% vs -11.84940%) is the T-bill overlay. Stress was the only module with a positive GFC total return (10.86351%).

4. The update: full sample and 2012–2026

The homework is not to re-print 1974–2011. It is to ask whether the same construction still does, after publication, what Antonacci said it did.

Table 4: Full sample 1998-11-30 to 2026-06-30 (332 months). Sharpe is arithmetic excess over T-bills. Trades/year is switches after the first holding.

Strategy	CAGR	Vol	Sharpe	Max DD	% + months	Trades/yr
US (VFINX)	9.02373%	15.16966%	0.51526	-50.96649%	64.46%	—
International (VGTSX)	6.32752%	16.79791%	0.33222	-58.48088%	59.64%	—
Equities relative	8.65837%	15.73172%	0.48132	-55.80419%	63.55%	1.19277
Equities dual	9.36744%	12.01473%	0.64161	-25.22119%	71.39%	1.48193
High yield	5.58654%	7.42720%	0.49871	-28.82840%	69.28%	—
Intermediate IG	4.64102%	5.25855%	0.50917	-18.46356%	64.76%	—
Credit relative	5.45173%	5.34252%	0.64351	-16.85414%	69.58%	1.19277
Credit dual	5.36550%	4.31775%	0.76409	-10.91063%	73.49%	1.66265
Equity REITs	8.85576%	20.11647%	0.42670	-68.27768%	61.14%	—
NLY (mREIT)	11.49515%	24.18115%	0.49661	-45.84467%	59.34%	—
REITs relative	15.99960%	19.76662%	0.75375	-44.86961%	62.35%	1.40964
REITs dual	10.16769%	17.43321%	0.53094	-52.22854%	65.06%	1.73494
Gold	10.08834%	13.24729%	0.64419	-39.27765%	53.01%	—
Long Treasuries	4.01670%	11.41052%	0.22664	-45.27119%	55.72%	—
Stress relative	9.80999%	12.81376%	0.64204	-24.48106%	57.53%	1.40964
Stress dual	9.06972%	12.26162%	0.60987	-24.48106%	59.04%	1.84337
1/8 risky, no momentum	8.30389%	9.30962%	0.69131	-26.47041%	66.87%	—
60/40	7.14834%	9.45346%	0.56702	-32.53137%	65.36%	—
T-bills	2.01736%	0.57661%	—	0.00000%	78.31%	—
Composite relative	10.60593%	9.11387%	0.93676	-24.97059%	65.66%	—
Composite absolute	6.87550%	5.39461%	0.89180	-8.30989%	67.77%	—
Composite dual (paper rule)	9.01638%	7.44579%	0.93231	-14.98158%	64.76%	—

Three facts from Table 4, relative to Antonacci's Table 10 / Table 14. First, dual still beats buy-and-hold *on risk*: composite vol 7.44579% vs US 15.16966% vs 1/8 9.30962%, max DD -14.98158% vs US -50.96649%. Second, dual no longer beats relative-only on return. Relative composite CAGR 10.60593% vs dual 9.01638%; Sharpes are almost the same (0.93676 vs 0.93231). The T-bill overlay bought a smaller left tail (DD -14.98158% vs -24.97059%) and paid for it in bull-market CAGR. Third, the REIT sleeve is where the paper and the update disagree most: relative REITs 15.99960% vs dual 10.16769%. In Table 6 of the paper those two were 16.80% and 16.78%. Here the cash overlay on NLY/VGSIX is expensive. That is partly NLY, partly the 2022–2023 REIT bear being slow enough for 12-month momentum to stay long and then getting whipped.

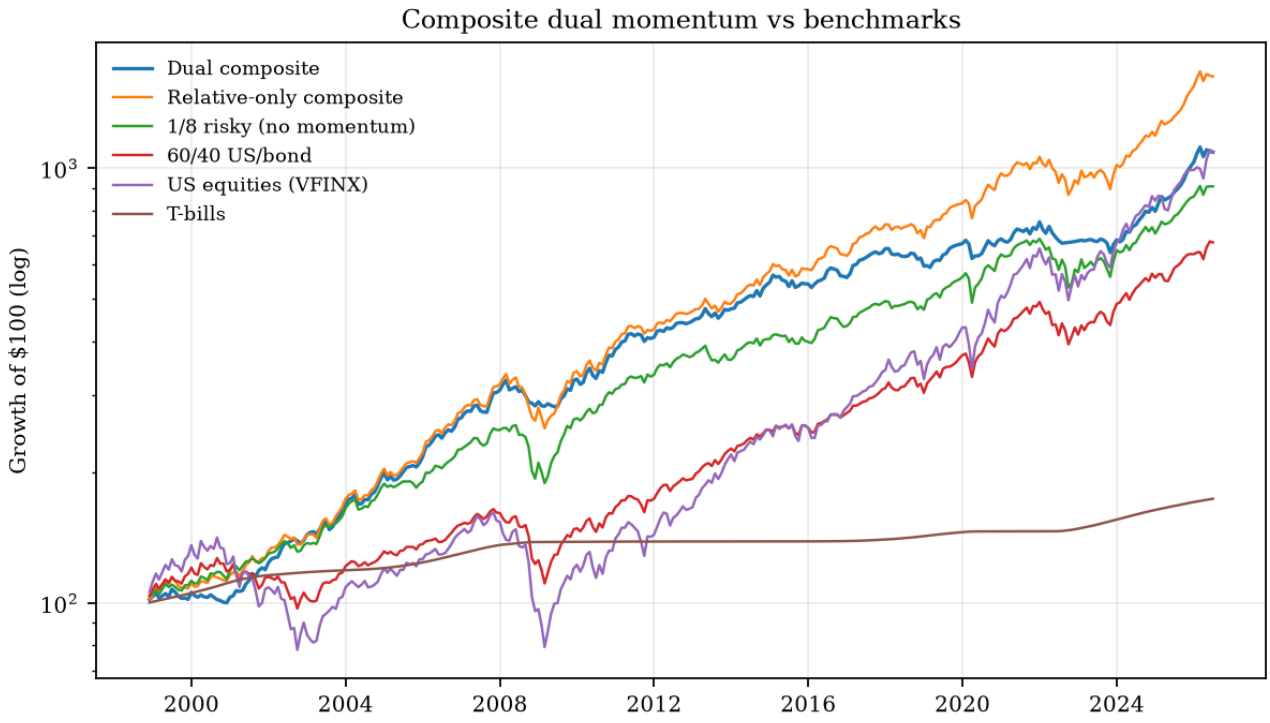


Figure 3: Growth of $100, log scale, full sample. Dual (blue) tracks relative (orange) through 2011, then lags as the T-bill overlay spends more time out of equities in a long bull market.

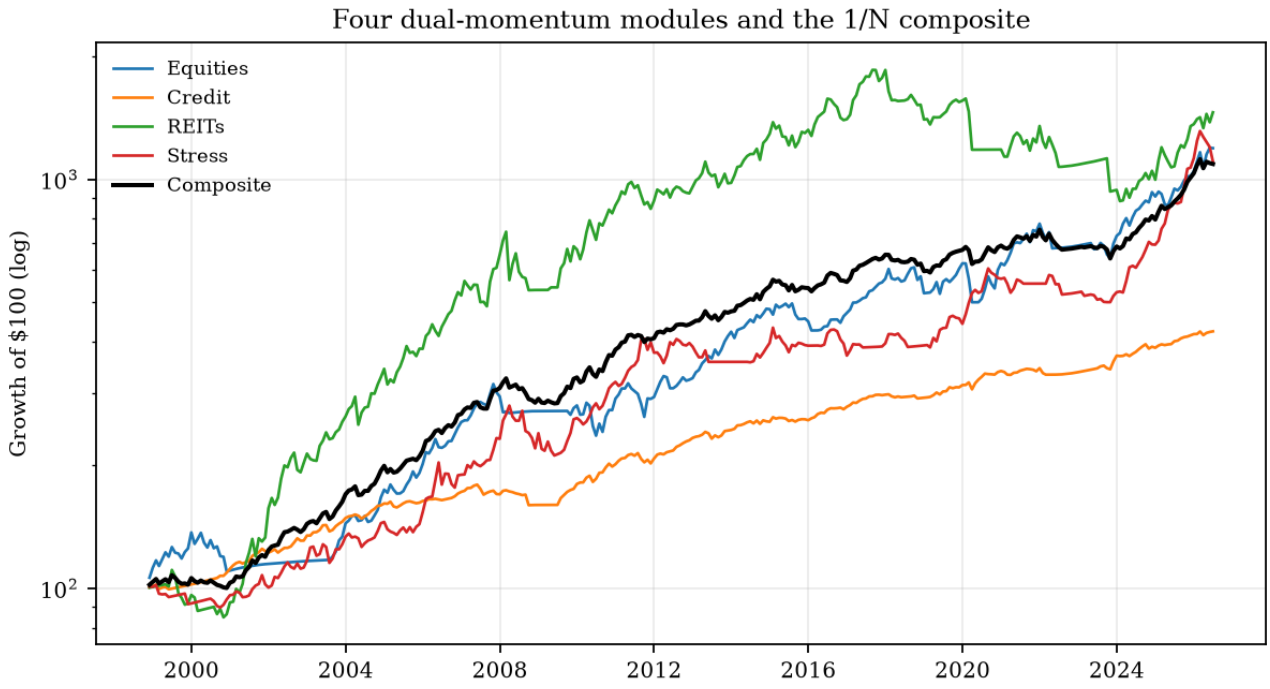


Figure 4: The four dual-momentum modules and the 1/N composite. Credit is the ballast; REITs and stress are the vol.

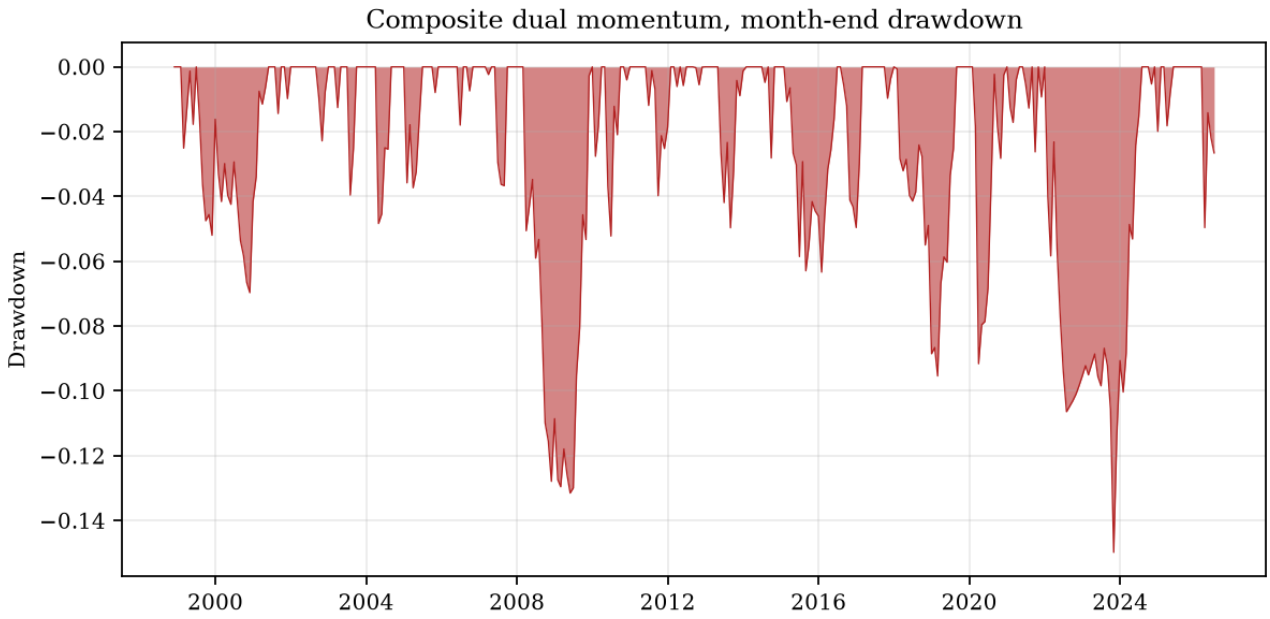


Figure 5: Composite month-end drawdown. The worst print is not the GFC; it is 2022–2024, when stocks, credit and REITs fell together and 12-month absolute momentum was late.

4.1 Out of sample, January 2012 – June 2026

Table 5: True out-of-sample window, Jan 2012 – Jun 2026 (174 months).

Strategy	CAGR	Vol	Sharpe	Max DD	% + months	Trades/yr
US (VFINX)	15.01906%	14.06593%	0.95940	-23.95956%	70.11%	—
60/40	9.81373%	9.17344%	0.89991	-20.16257%	71.26%	—
1/8 risky	7.12343%	9.72732%	0.59859	-22.79939%	66.67%	—
Equities dual	10.14698%	12.15368%	0.73070	-19.66101%	70.11%	2.00000
Credit dual	5.06484%	3.82034%	0.90135	-3.79864%	71.84%	1.79310
REITs dual	3.49395%	15.48815%	0.19944	-52.22854%	64.94%	2.27586
Stress dual	7.70246%	11.14573%	0.58711	-17.27276%	52.87%	2.06897
Composite relative	9.68081%	9.12155%	0.89313	-18.21409%	64.37%	—
Composite dual	6.99688%	7.25928%	0.75767	-14.98158%	63.22%	—
Composite 12-1 (extra credit)	9.00196%	7.61128%	0.97598	-12.70662%	68.97%	—

This is the result that should change how one reads the paper. US equities compounded at 15.01906% with Sharpe 0.95940. Dual composite compounded at 6.99688% with Sharpe 0.75767. 60/40 made 9.81373%. Relative-only composite made 9.68081% — so the drag is not “momentum died,” it is specifically the *absolute* overlay in a decade-plus bull with short, sharp drawdowns. Split the OOS in two: 2012–2019 dual CAGR 6.40723% vs US 14.72234%; 2020–2026 H1 dual 7.72709% vs US 15.38531%. Both halves, same story.

Module-level OOS is not uniform. Credit dual still looks like the paper: CAGR 5.06484%, vol 3.82034%, Sharpe 0.90135, max DD only -3.79864%. REIT dual is the failure: CAGR 3.49395%, max DD -52.22854%. Equities dual 10.14698% vs buy-and-hold US 15.01906% is the cost of being in T-bills during 2015–16, 2018, 2022 and parts of 2022–23 while the 12-month window reset. Stress dual 7.70246% beat long Treasuries (0.75516%) by sitting in gold, which is the module working.

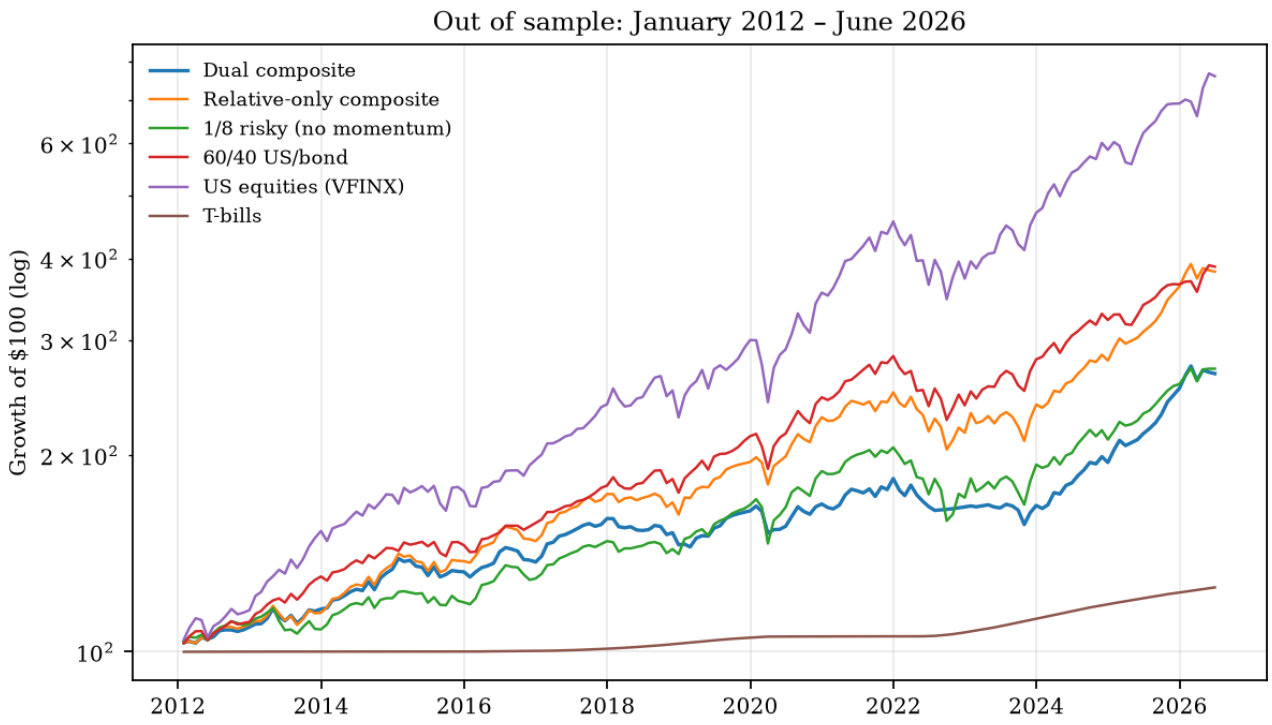


Figure 6: OOS growth of $100. US equities (purple) pull away. Dual (blue) ends near the 1/8 risky portfolio and below 60/40 — the opposite of Table 10's ranking.

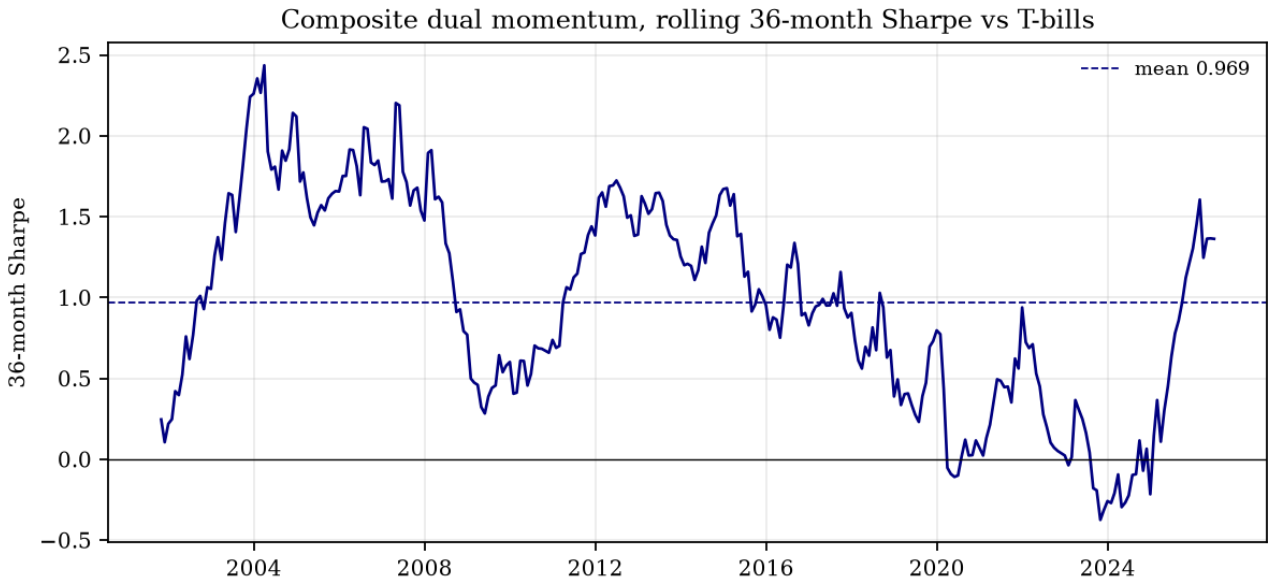


Figure 7: Rolling 36-month Sharpe of the dual composite vs T-bills. The post-publication window is a sequence of near-zero and negative prints around 2015, 2018 and 2022, not a stable 1.0.

4.2 Three crises the 12-month rule was and was not built for

Table 6: Crisis windows, total return (not annualized). Dual and relative are identical in COVID because 12-month absolute momentum was still positive.

Window (total return)	Dual	Relative	EW8	60/40	US
GFC Oct 07 – Mar 09	-0.91400%	-11.84940%	-19.87949%	-27.63502%	-45.82052%
COVID Feb–Mar 2020	-9.16065%	-9.16065%	-14.46438%	-11.61005%	-19.62056%
2022 hiking cycle	-9.53777%	-13.25944%	-16.22648%	-15.95193%	-18.24441%

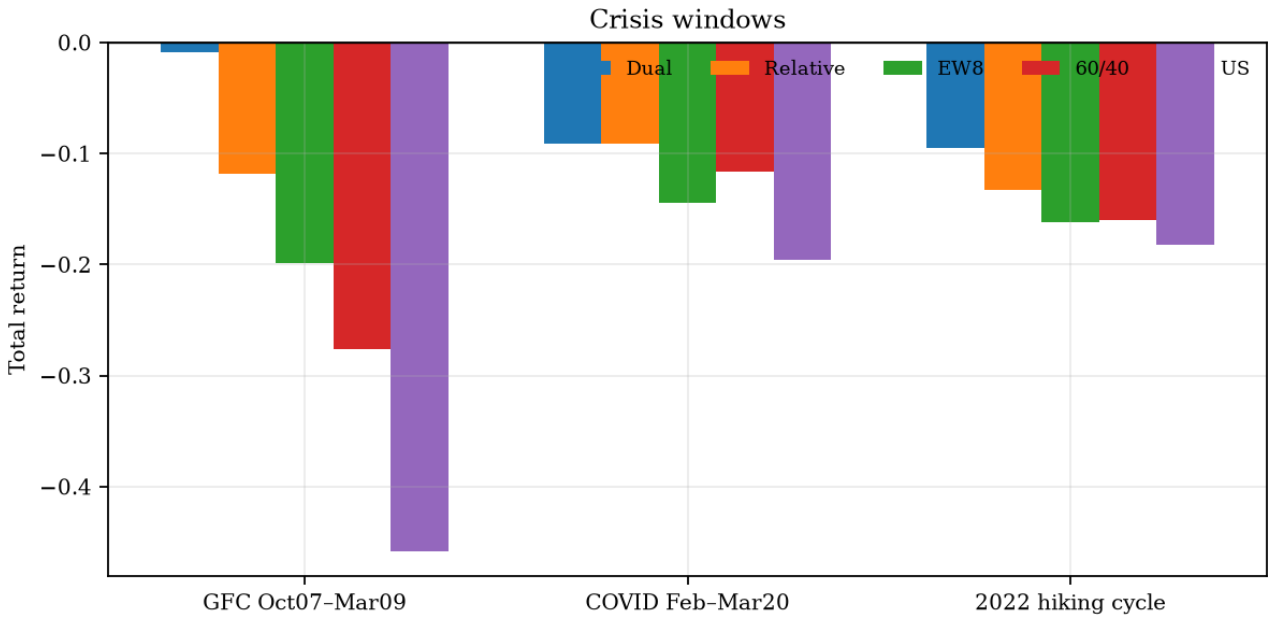


Figure 8: Same crisis windows as Table 6.

GFC: the textbook case. A long, deep, correlated risk-off that 12-month absolute momentum can see. COVID: the opposite. The crash is two months inside a still-positive 12-month window, so dual = relative = long equities and REITs, and the composite drops -9.16065% while US drops -19.62056%. A 12-month timer cannot help you in March 2020; a 3-month timer might, and Table 9 of the paper already showed that 3-month formation is worse on average. 2022: stocks and bonds fall together, so 60/40 does not diversify. Dual loses less than 60/40 (-9.53777% vs -15.95193%) because credit goes to T-bills and stress prefers gold to Treasuries, but REITs still lose -19.51149% and the composite's worst drawdown of the whole sample is this episode plus the 2023–24 hangover, not 2008.

4.3 What I would actually conclude

Antonacci's qualitative claims that survive 2012–2026: (i) modules keep you from putting the whole book in one correlated winner; (ii) absolute momentum cuts left-tail damage in slow bears (GFC, parts of 2022); (iii) credit dual momentum is still a high Sharpe, low DD sleeve. The quantitative claim that does not survive: that combining relative and absolute momentum raises return *and* Sharpe relative to the assets and to 1/N (Tables 10 and 14). After publication, in public funds, dual is a risk overlay on a relative-momentum book. It is not an alpha machine vs 60/40 or vs US equities. Anyone running this live in 2012–2026 underperformed a lazy 60/40 on CAGR and only roughly matched it on Sharpe (0.75767 vs 0.89991).

5. Extra credit: 12-1 lookback and inverse-vol weights

Two changes, specified in the plan before OOS numbers were computed. No lookback search, no in-sample tuning of the vol window.

12-1. Same dual rule, but $m_{i,t}$ uses months *t*−12 through *t*−1, skipping month *t*. This is the Jegadeesh–Titman / AQR convention the paper discussed and rejected for mixed assets, and it is the reading of the assignment's "without last-month reversal effects."

Inverse-vol 1/N replacement. At t, estimate each module's trailing 12-month realized vol from that module's dual returns through t, then hold weights $\propto 1/\sigma$ during t+1. No caps. This asks whether 25/25/25/25 is leaving Sharpe on the table because Table 10's stress sleeve has ~2× credit vol.

Table 7: Extra-credit horse race, frozen specifications, same assets, zero costs.

Construction, OOS 2012–2026 H1	CAGR	Vol	Sharpe	Max DD	% + months	Trades/yr
Dual 12-0, 1/N (paper)	6.99688%	7.25928%	0.75767	-14.98158%	63.22%	—
Dual 12-1, 1/N	9.00196%	7.61128%	0.97598	-12.70662%	68.97%	—
Dual 12-0, inv. vol	5.99819%	6.74146%	0.66607	-18.99989%	69.36%	—
Dual 12-1, inv. vol	8.12312%	6.66381%	0.98314	-10.74428%	71.68%	—
Relative 12-0, 1/N	9.68081%	9.12155%	0.89313	-18.21409%	64.37%	—

Table 8: Same horse race on the full sample (12-1 and inv-vol start a few months later).

Construction, full sample	CAGR	Vol	Sharpe	Max DD	% + months	Trades/yr
Dual 12-0, 1/N	9.01638%	7.44579%	0.93231	-14.98158%	64.76%	—
Dual 12-1, 1/N	9.67321%	7.59537%	0.99754	-13.56795%	67.37%	—
Dual 12-0, inv. vol	6.97608%	6.24402%	0.80797	-18.99989%	70.85%	—
Dual 12-1, inv. vol	7.92549%	6.26733%	0.95359	-12.32724%	72.96%	—
Relative 12-0, 1/N	10.60593%	9.11387%	0.93676	-24.97059%	65.66%	—

Skip-a-month wins, out of sample and in sample. OOS CAGR rises from 6.99688% to 9.00196%, Sharpe from 0.75767 to 0.97598, max DD shrinks from -14.98158% to -12.70662%. Full-sample Sharpe goes from 0.93231 to 0.99754. Last-month reversal is not just an equity microstructure story; in this mixed-asset book, including month t in a 12-month rank that is then held during t+1 adds noise. The paper's consistency argument for not skipping is reasonable and I kept it as the baseline. The data after 2011 disagree with it.

Inverse-vol loses on the thing the paper was designed to do. Mean weights: credit 46.851%, equities 24.438%, stress 17.209%, REITs 11.556%. That is mechanically correct — credit vol is ~4% — and it is the opposite of Antonacci's premise that you *want* the high-vol sleeves, with a trend overlay, because that is where the risk premium lives. OOS inv-vol CAGR 5.99819% with a left tail that is actually worse than 1/N in 2020 (kurtosis 21.80552 Pearson). Combining 12-1 with inverse-vol repairs some of the DD (-10.74428% OOS) but still trails 12-1 1/N on CAGR. I would not take it.

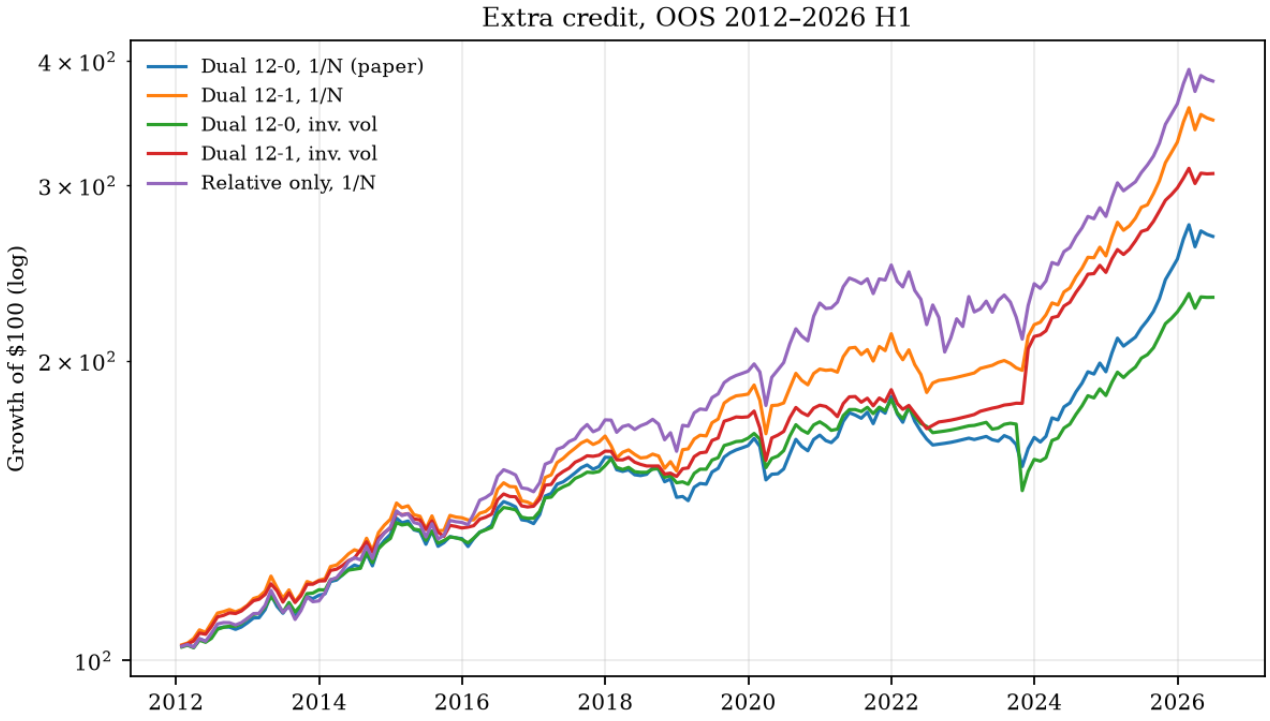


Figure 9: Extra credit, OOS. 12-1 (orange) and relative-only (purple) pull away from the paper rule (blue). Inverse-vol (green) is the laggard.

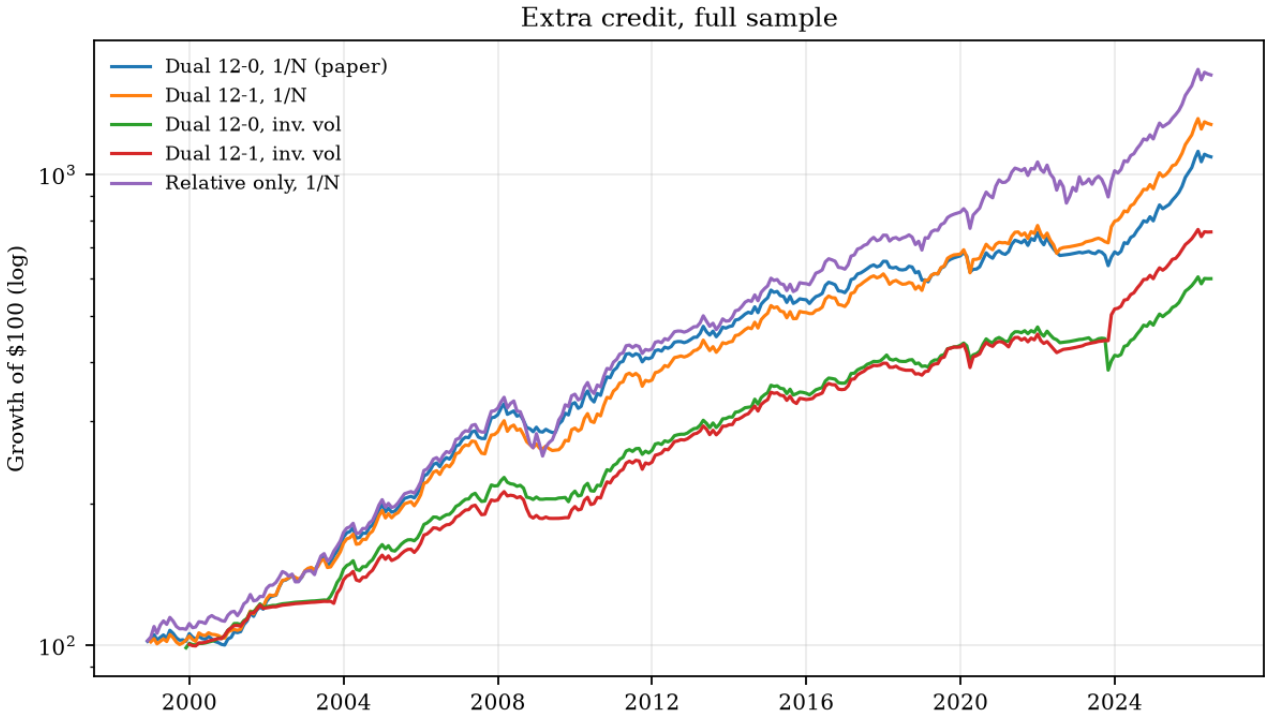


Figure 10: Extra credit, full sample. Same ranking.

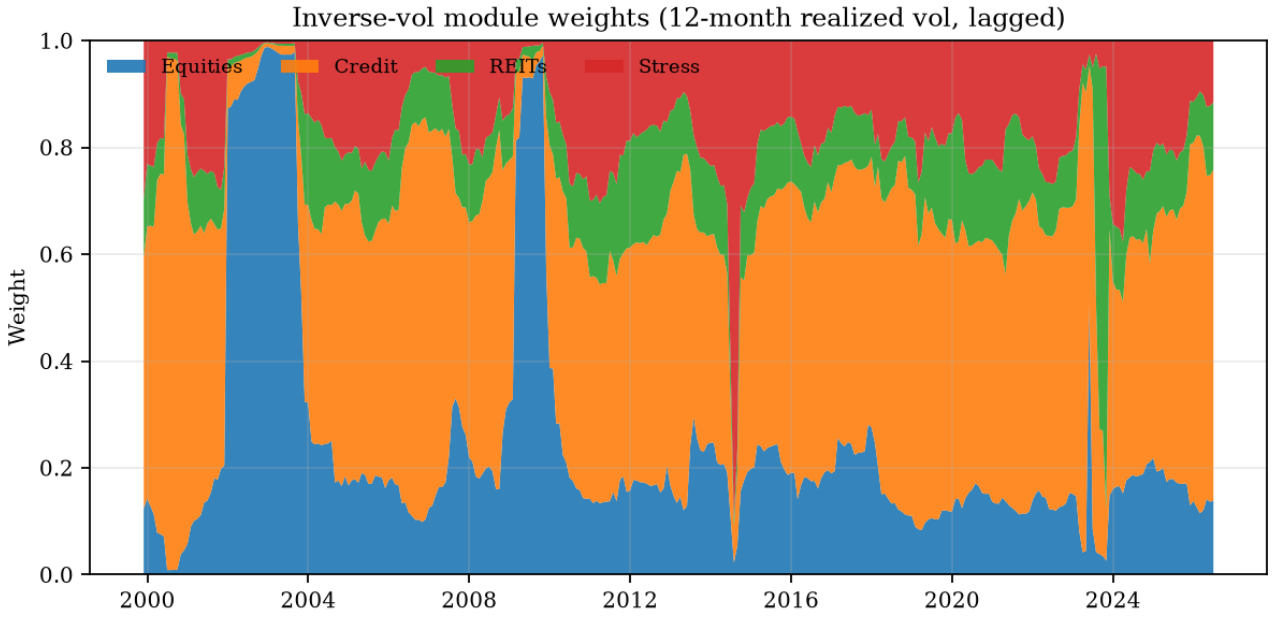


Figure 11: Inverse-vol weights over time. Credit (second band) is structurally overweight; REIT and stress are structurally underweight. That is why inv-vol fights the strategy.

If I had to ship one change: switch the baseline lookback from 12-0 to 12-1, keep 1/N modules, keep the T-bill hurdle. Do not risk-parity the modules.

6. Limitations

Live funds, not the paper's indices: Vanguard expense ratios and NLY's single-name risk sit in the returns. That biases us *against* matching Table 10, which is the conservative direction for an update. Gold is a monthly average, not a PM fix. No transaction costs; paper turnover is 1.2–1.8 switches per module-year here too, so costs are small relative to the OOS gap vs US equities. 60/40 uses US bonds, not the paper's world 60/40. The composite is not volatility-targeted at the asset level. I did not walk the strategy through a capacity or tax overlay. French RF is a 1-month bill. None of these explain away the OOS underperformance vs US equities; they do mean Table 2 is a cousin of Table 8, not a clone.

7. Reproducibility

Python 3.12, from `hw5/`:

```
python3.12 -m venv .venv && .venv/bin/pip install -r requirements.txt
     MPLCONFIGDIR=.mplcache .venv/bin/python -m pytest tests/ -q
      MPLCONFIGDIR=.mplcache .venv/bin/python scripts/run_all.py
   MPLCONFIGDIR=.mplcache .venv/bin/python scripts/build_report.py
```

Network is required only for a fresh download (`python scripts/run_all.py --download`). The graded path uses `data/processed/monthly_returns.csv`. Timing convention, lookbacks and inverse-vol lag are unit-tested. Paper PDF: `docs/antonacci_2013_dual_momentum.pdf`.